HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Designed and deployed **machine learning models in production environments** using **Python (scikit-learn, pandas, NumPy)** and **R** for statistical modelling, including **OLS, multinomial logistic regression, and clustering algorithms**, achieving **92% model accuracy** on payment risk segmentation tasks across HCL's banking payments platform.
- Cleansed, structured, and managed **large and complex datasets** from enterprise payment systems using **SQL (PostgreSQL, MySQL, Oracle)** and **Python ETL pipelines**, reducing data preparation time by **40%** and enabling downstream analytics for cross-border payments and SEPA transaction flows.
- Built and maintained **Power BI dashboards and reports** tracking payment processing trends, transaction volumes, and operational KPIs for senior stakeholders; created **DAX measures** and **Power Query transformations** to validate business requirements and deliver **15+ executive-level insights monthly** on payments operations.
- Conducted **exploratory data analysis** and applied **advanced statistical techniques** (hypothesis testing, time-series analysis, A/B testing) on payment messaging data (**ISO 20022, SWIFT MT/MX formats**) to identify transaction anomalies and operational inefficiencies, presenting findings to banking domain teams and regulatory stakeholders.
- Processed large-scale data using **big data technologies** including **Spark and Hadoop** concepts, integrated **AWS (EC2, S3, Lambda)** for scalable analytics infrastructure, and leveraged **Docker containerisation** for reproducible machine learning pipelines serving **500K+ daily payment transactions**.
- Collaborated with **technical and non-technical stakeholders**—including banking client teams, business analysts, and operations staff—to translate analytical findings into actionable business outcomes; supported UAT, change-control processes, and post-implementation triage, ensuring delivery of data-driven solutions within agreed timelines and budget constraints.

SKILLS

Data Science & Machine Learning – Python (scikit-learn, pandas, NumPy), R, TensorFlow, PyTorch, OLS regression, multinomial logistic regression, LDA, clustering, segmentation, model validation, supervised and unsupervised learning.

Data Engineering & Analytics – SQL (PostgreSQL, MySQL, Oracle), data cleansing, data modelling, data warehousing, ETL pipelines, exploratory data analysis, statistical analysis, hypothesis testing, time-series analysis.

Business Intelligence & Visualization – Power BI, Tableau, DAX, Power Query, dashboard design, KPI tracking, business reporting, data visualisation.

Big Data & Cloud Infrastructure – Spark, Hadoop, AWS (EC2, S3, Lambda, CloudWatch), Docker, Git, Jenkins, scalable data processing.

Payments & Domain Expertise – SWIFT messaging (MT/MX formats), ISO 20022 message flows, SEPA payments, cross-border payments, PSD2, Open Banking, payment processing workflows, payments operations dashboards, regulatory compliance (GDPR, data protection).

AI & Advanced Analytics – Artificial Intelligence, Generative AI, Agentic AI concepts, machine learning frameworks, statistical modelling, advanced analytics.

Professional Skills – Stakeholder communication, translating technical concepts for non-technical audiences, Agile/Scrum, change-control management, UAT coordination, cross-functional collaboration, project delivery.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

AI-Driven Payment Risk Segmentation & Fraud Detection Platform (HCL Technologies)

- Built end-to-end **machine learning pipeline** using **Python (scikit-learn, pandas)** and **R statistical modelling** to segment payment transactions into risk cohorts via **multinomial logistic regression, clustering, and LDA**, processing **12M+ monthly payment records** sourced from **PostgreSQL and Oracle** payment systems.
- Constructed scalable data infrastructure using **Spark** for **large-scale data processing** and containerised the ML models with **Docker**, deployed on **AWS Lambda** for real-time inference; achieved **94% precision on high-risk transaction detection** and reduced false-positive alerts by **38%**.
- Designed **Power BI dashboards** with **DAX measures** to visualise model predictions, transaction risk scores, and operational trends; created **Power Query transformations** for automated data refresh and delivered **weekly stakeholder reports** on payment risk metrics and compliance status.
- Applied **exploratory data analysis** and **statistical hypothesis testing** to validate model assumptions across **ISO 20022 and SWIFT payment message flows**; collaborated with banking domain teams and regulatory stakeholders to ensure segmentation logic aligned with SEPA compliance and cross-border payment regulations.
- Supported **production monitoring** and **model validation** using **AWS CloudWatch**; implemented automated retraining pipelines and established SLA-driven alerting for model drift detection, ensuring **99.8% system uptime** across **500K+ daily payment transactions**.

SWIFT Payments Operations Dashboard & Real-Time Alerting (HCL Technologies)

- Engineered **Power BI dashboard** consolidating **SWIFT MT/MX message data** and **ISO 20022 transaction flows** for banking payments operations; applied **SQL queries** across **PostgreSQL and Oracle** for real-time transaction-flow visibility and implemented **DAX-based KPI measures** for SLA tracking and regulatory reporting.
- Processed and validated **large, complex datasets** from payment messaging systems using **Python (pandas, NumPy)** and **SQL data manipulation**; cleansed **2.5M+ daily payment records** and structured them for downstream analytics, reducing data validation errors by **35%** and enabling accurate reconciliation workflows.
- Delivered **executive-level insights** on payment processing trends, cross-border payment latency, and SEPA compliance metrics to senior banking stakeholders; supported UAT coordination and post-implementation triage with clear communication of analytical findings and business impact.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)